

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Algebra	Linear equations in two variables	Easy

Question

In a chess tournament, each participant earns **1** point for each game the participant plays that ends in a draw and **3** points for each game the participant wins. A certain participant in this tournament has earned **41** points. Which equation represents this situation, where *d* represents the number of games this participant has played that ended in a draw and *w* represents the number of games this participant has won?

Answer

- A. $d + 3w = 41$
- B. $3d + w = 41$
- C. $d + \frac{w}{3} = 41$
- D. $\frac{d}{3} + w = 41$

Correct Answer: A

Rationale

Choice A is correct. It's given that each participant earns **1** point for each game the participant plays that ends in a draw and **3** points for each game the participant wins. Since *d* represents the number of games that a certain participant has played that ended in a draw and *w* represents the number of games this participant has won, this participant has earned a total of **1d + 3w**, or **d + 3w**, points. Since this participant earned **41** points, the equation **d + 3w = 41** represents this situation.

Choice B is incorrect. This equation represents the situation where the participant earned **3** points for each game played that ended in a draw and **1** point for each game won and earned **41** points.

Choice C is incorrect. This equation represents the situation where the participant earned **1** point for each game played that ended in a draw and $\frac{1}{3}$ point for each game won and earned **41** points.

Choice D is incorrect. This equation represents the situation where the participant earned $\frac{1}{3}$ point for each game played that ended in a draw and **1** point for each game won and earned **41** points.